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Global Quality Solutions

Jan–Sep 2026

Non-Rayong overview 2026.

Service hours, site mix and customers, then cost reduction and operation excellence, across our three non-Rayong sites.

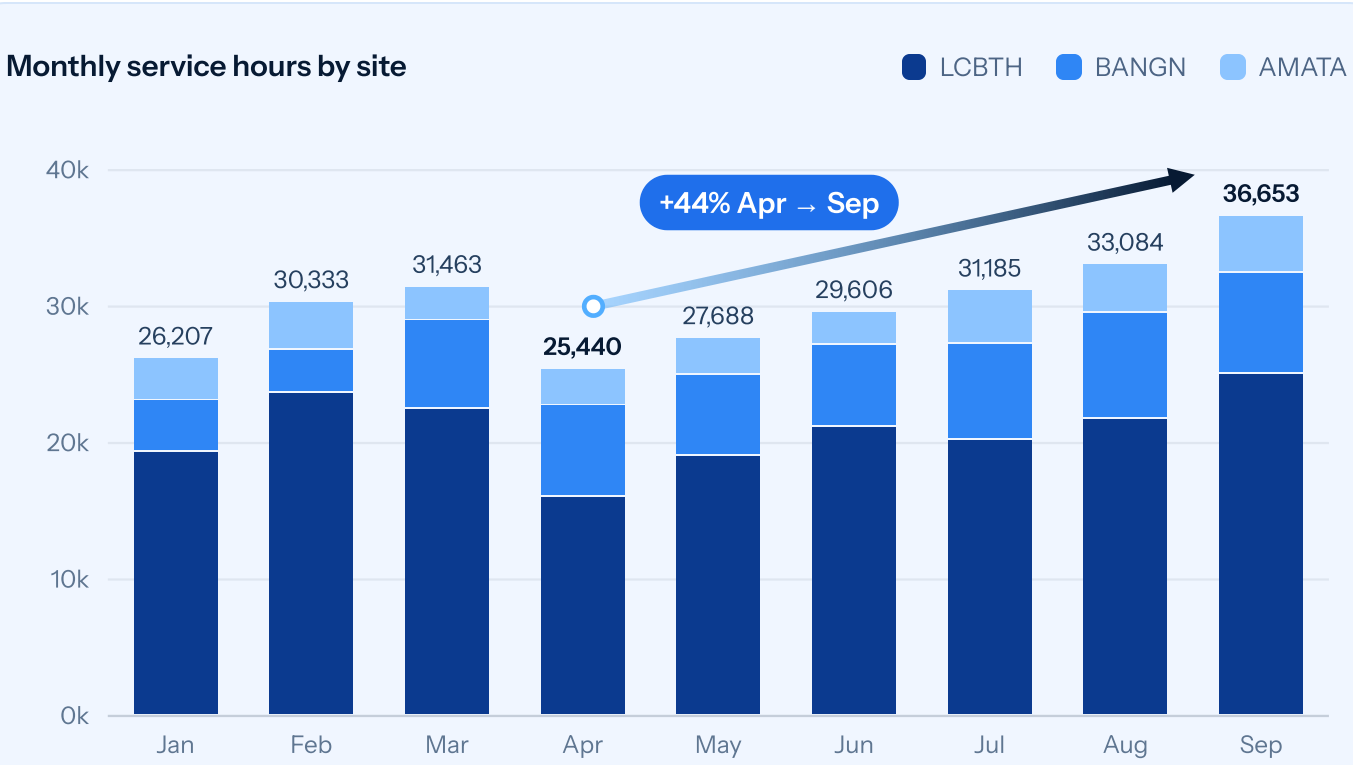
LCBTH BANGN AMATA

TRIGO Quality Services (Thailand)

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01 · Performance

September set a record after the April dip.



−19%

Apr low vs Mar, 25,440 h; LCBTH −29%

+44%

Apr to Sep: 25,440 h to a record 36,653 h

176

FTE in Sep vs 141 Jan–Aug avg

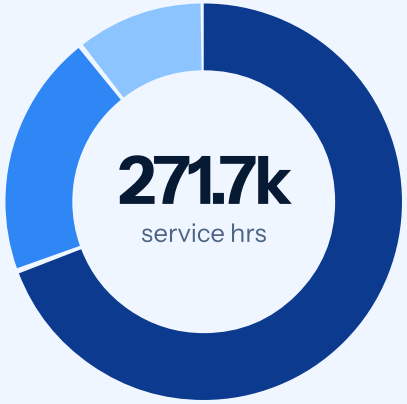
Quarterly service hours

Q1	88.0K h
Q2	82.7K h
Q3	**100.9K h**

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01 · Site mix · Jan–Sep 2026

LCBTH leads hours; BANGN leads missions.



LCBTH	69.5%
BANGN	20.0%
AMATA	10.6%

LCBTH

Service hours	188,739
Share of hours	69.5%
Customers	20
Missions	76
Hours / mission	2,483
Mgmt fee days	2,004
Per-piece units	5,985
Sep vs Jan hours	+30%

LCBTH = scale

Ford + AutoAlliance + Mitsubishi port/plant inspection. ~2,500 hrs per mission; only 10 hosts.

BANGN

Service hours	54,235
Share of hours	20.0%
Customers	53
Missions	140
Hours / mission	387
Mgmt fee days	1,200
Per-piece units	23,533
Sep vs Jan hours	+96%

BANGN = breadth

140 missions, 53 customers, 70% of per-piece units. Many short sorting jobs (~390 hrs/mission).

AMATA

Service hours	28,687
Share of hours	10.6%
Customers	31
Missions	45
Hours / mission	637
Mgmt fee days	663
Per-piece units	4,160
Sep vs Jan hours	+35%

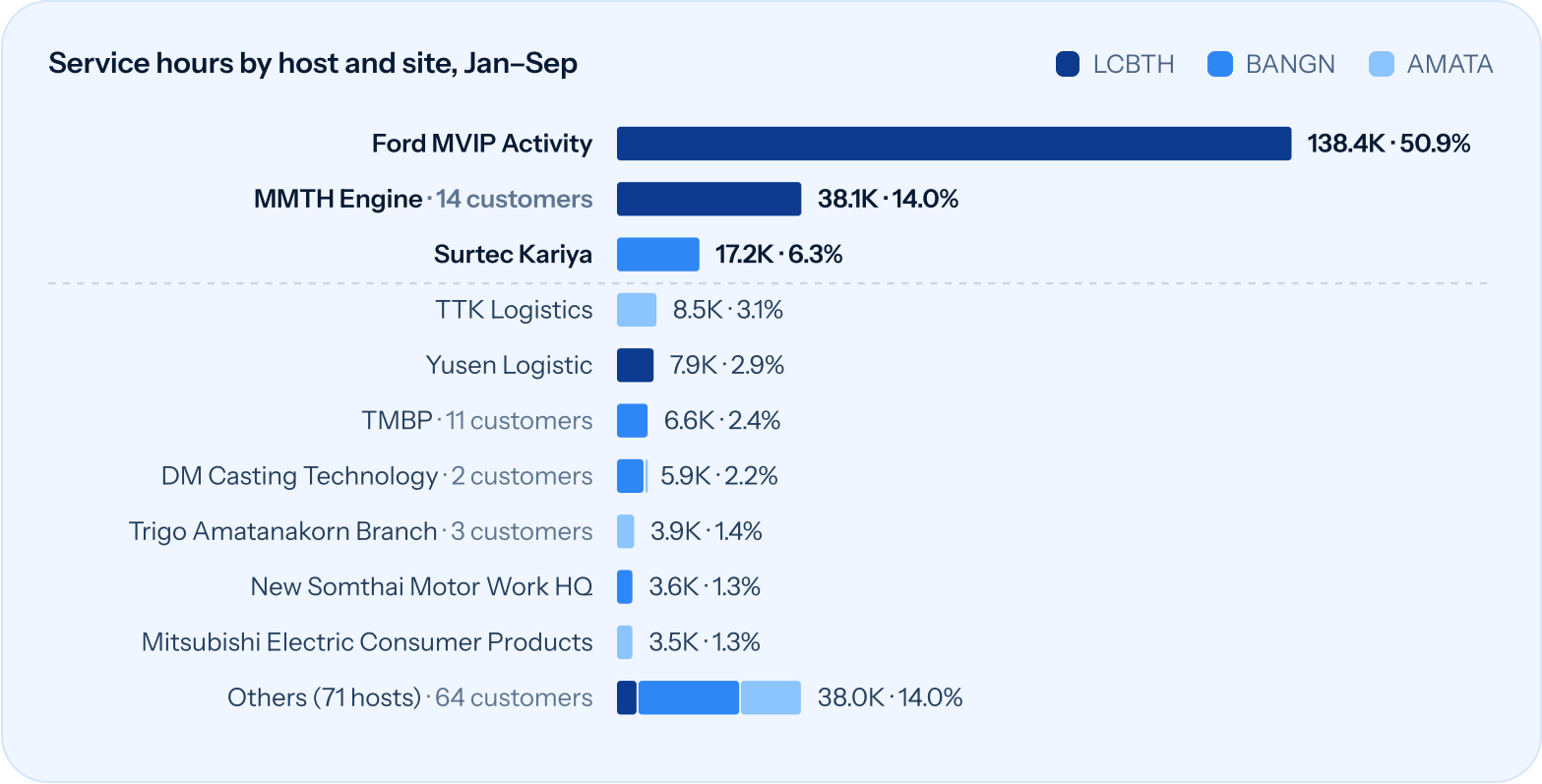
BANGN = growth

Hours up 96% from Jan to Sep, the fastest growth of the three sites.

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01 · Customers & hosts

Three major hosts account for 71% of service hours.



51%

Ford MVIP activity: 138.4K h across the Ford and Auto Alliance plants

14

Customers at MMTH Engine: largest host, 38.1K h; TMBP serves 11

4.5×

Yusen Logistic growth: 331 h in Jan → 1,488 h in Sep

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Non-Rayong overview 2026

Cost reduction.

A new beverage supplier, and a 2026 drink mix that lowers the cost per unit.

02

02 · Beverage supplier

Supplier change reduced unit cost by 30%.

Lower purchase cost, with the monthly operating trend kept in view.

PREVIOUS SUPPLIER

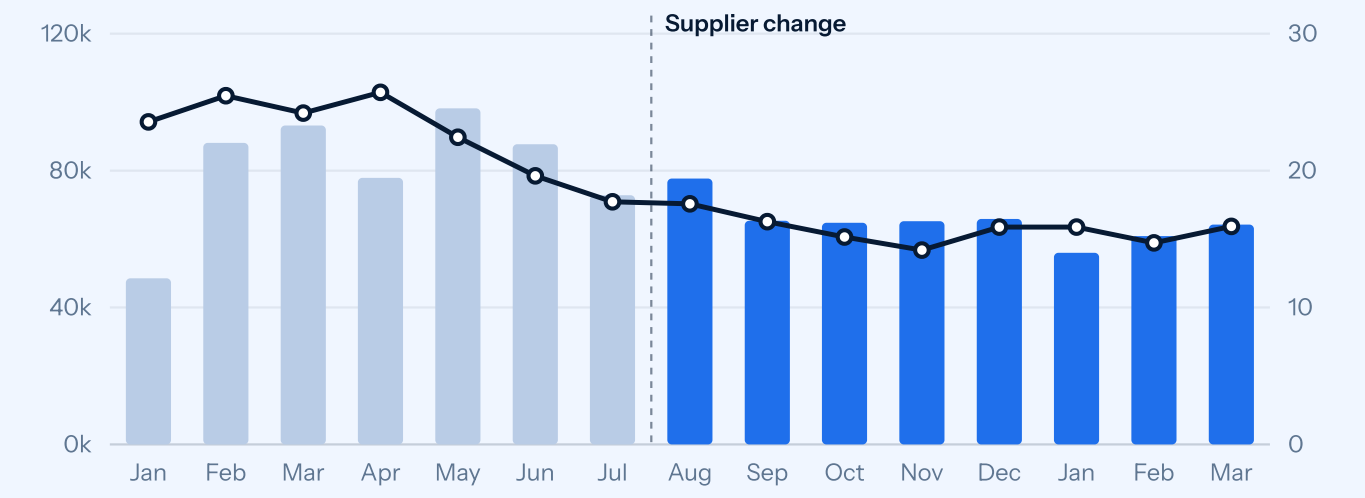
22.34 THB / unit

Monthly avg. 80,899.75 THB

CURRENT SUPPLIER

15.68 THB / unit

Monthly avg. 64,983.03 THB



Monthly sequence as supplied · Supplier change from Aug onward

02 · Drink mix

2026 drink mix targets 47.5% lower unit cost.

Redistribute the beverage mix and introduce herbal drinks and tea.

BEVERAGE ALLOCATION	Y2025	Y2026
Electrolyte drink	21%	50%
Soft drink	53%	14%
Energy drink	21%	15%
Ready to drink coffee	8%	7%
Herbal drink New	-	7%
Tea New	-	7%

EXPECTED COST / UNIT

11.73

THB / unit

10.61 THB saved per unit

47.5% below 2025

Source check required: 2025 electrolyte allocation is 21% in the table but 18% in the source pie. 2025 table totals 103%; all supplied allocations are retained pending confirmation.

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Non-Rayong overview 2026

Operation excellence.

Handheld OCR, a hybrid online and offline workflow, and four poka-yoke controls.

03

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03 · Handheld OCR

Handheld OCR replaces manual serial-number recording.

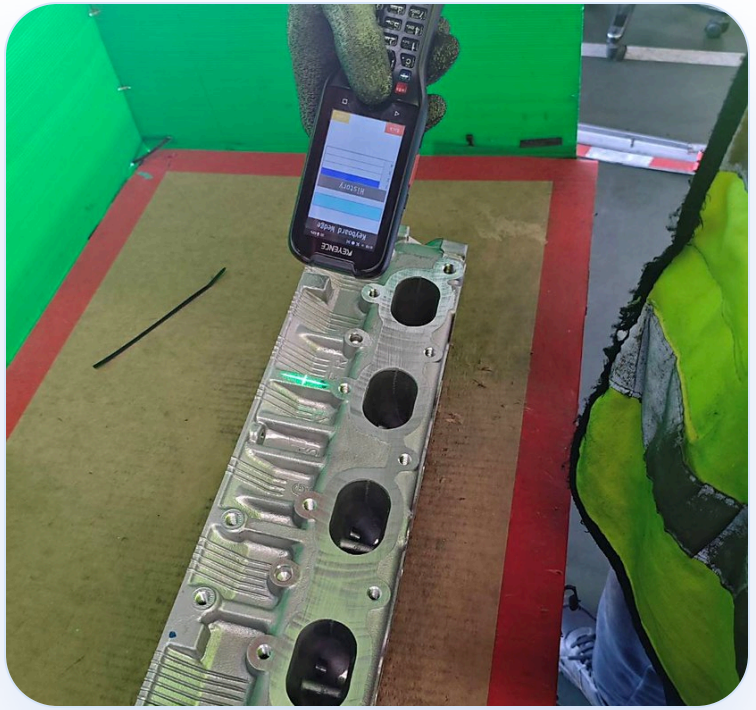
Read part serials with handheld OCR and automatically record inspection data in Google Sheets.

8–10 sec MANUAL → **1 sec** OCR + AUTO RECORD

−87.5–90% TIME PER SERIAL

MISSION	MARKING	MANUAL	OCR + AUTO RECORD	TIME SAVED
LCBTH.26.0005	Dot peen	8 sec	1 sec	−87.5%
LCBTH.26.0002	Dot peen	8 sec	1 sec	−87.5%
LCBTH.26.0027	Stamped	10 sec	1 sec	−90%
LCBTH.26.0007	Embossed	9 sec	1 sec	−88.9%

Dot peen / stamped / embossed · Barcode / Data Matrix · Internal configuration; supplied mission comparisons.





Shopfloor OCR in use

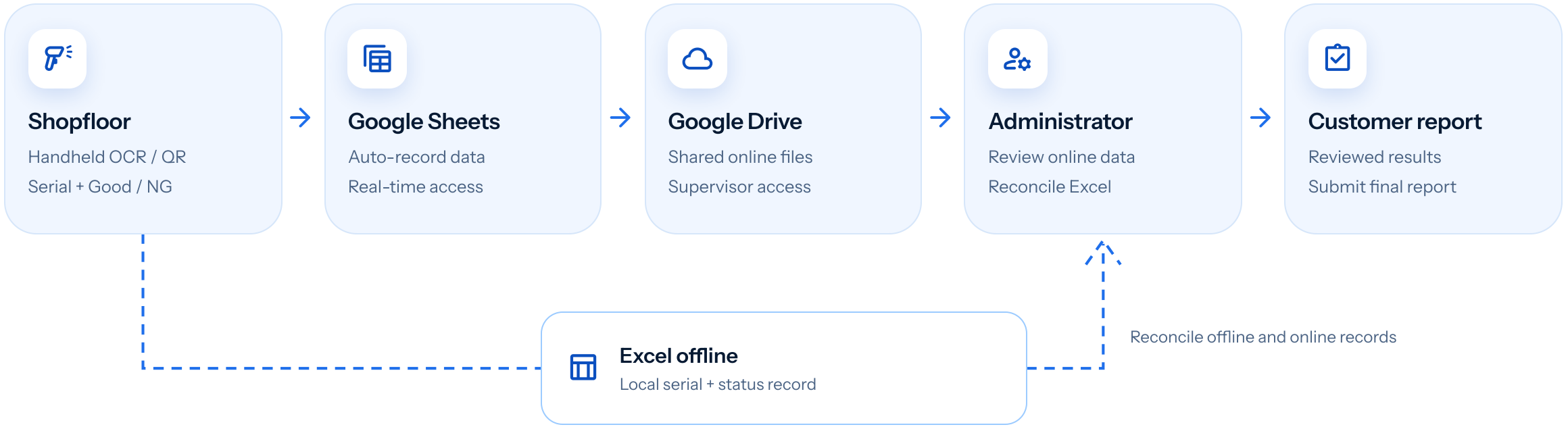
KEYENCE BT-A1000 · LCBTH.26.0005

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03 · Hybrid workflow

A hybrid workflow connects shopfloor capture to customer reports.

Online cloud records + offline Excel support → administrator review → customer report.



Source: supplied OCR capture and MMTH hybrid workflow. Generic icons denote functions, not product logos.

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03 · Poka-yoke

Four controls protect every inspection record.

Visual + voice guidance at capture. Administrator review before customer reporting.

Wrong part / serial

Validate against customer list / production plan. Warn on mismatch.

Duplicate scan

Lock out repeated scans. Prevent duplicate counting.

Full box

Voice + popup notification when target quantity is reached.

Review & release

Administrator checks offline + online records before submitting the customer report.

Linked data: Customer · Material · Batch · Order | Missions 04, 29, 31 · Internal configuration + Text-to-Speech support shopfloor use.

Source: supplied error-prevention workflow. Next: two web apps supporting Non-Rayong operations.

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Non-Rayong overview 2026

The two apps.

The Manpower Board plans who goes where. The Inventory & Quality System tracks tools, parts and consumables.

04

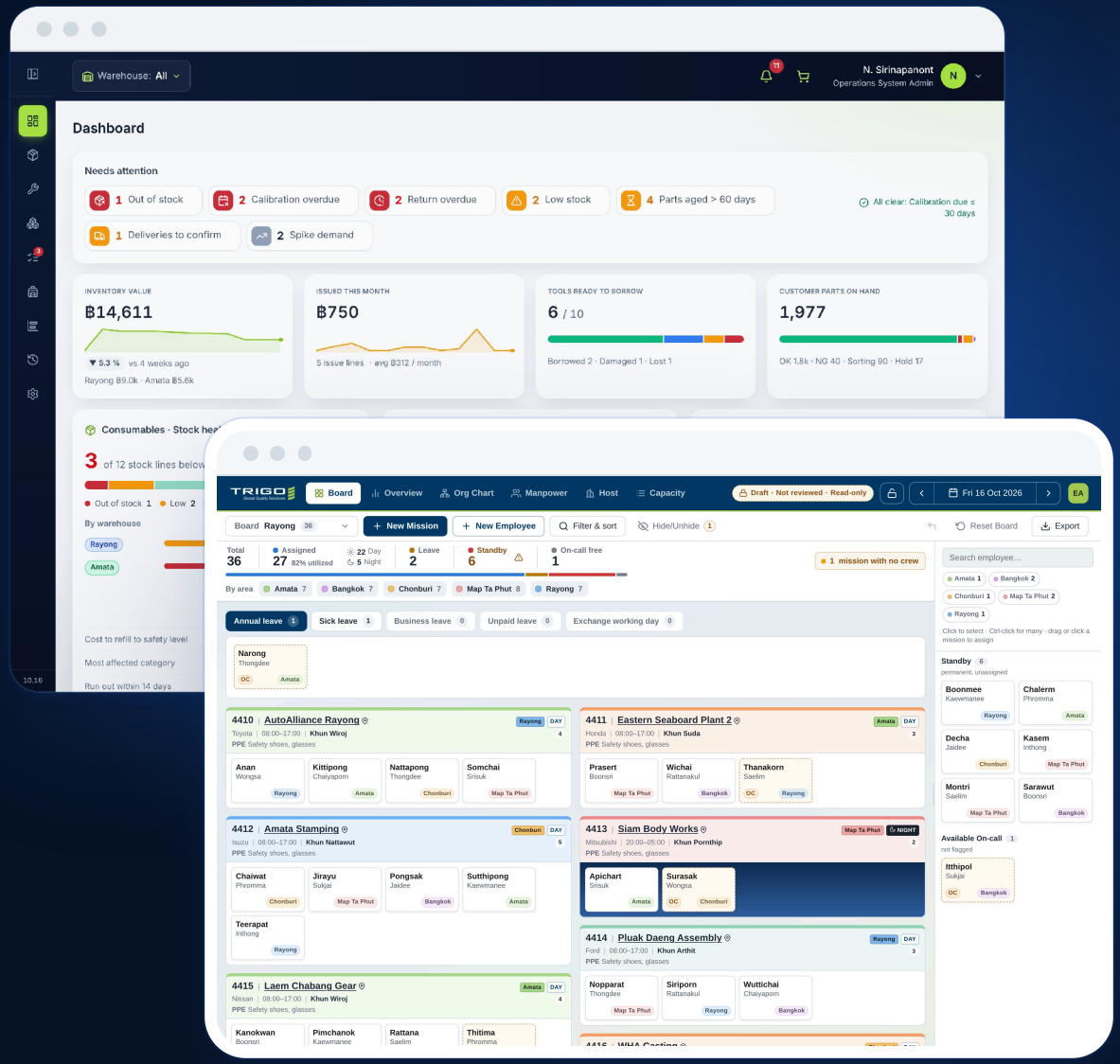
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04 · The two apps

Two apps for the work we do every day.

The Manpower Board plans who goes where. The Inventory & Quality System tracks tools, parts and consumables. Both run in the browser and on your phone.

Manpower Board | Inventory & Quality | On your phone



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Manpower Board

04 · App 1 of 2

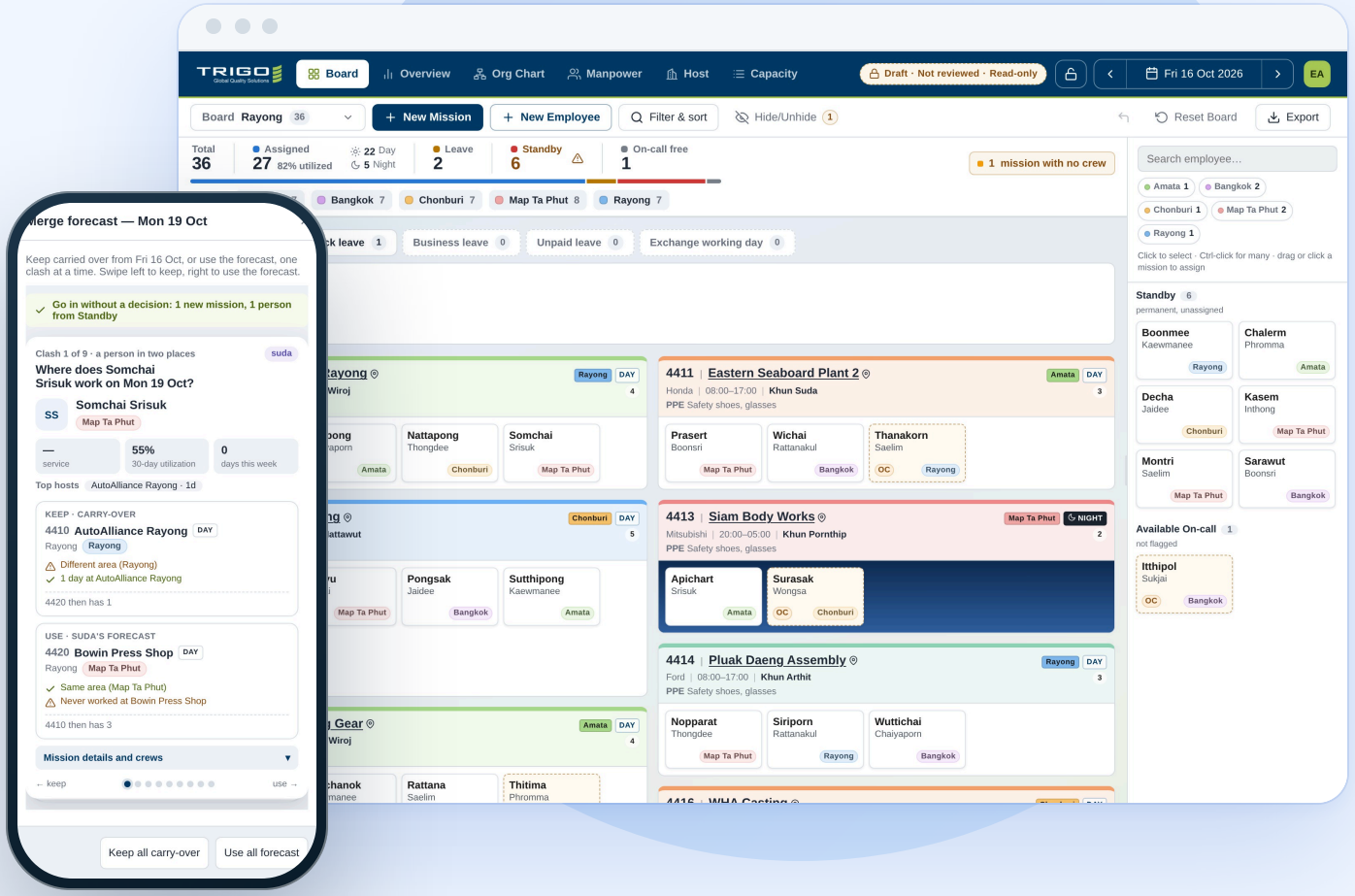
Tomorrow's crew, planned on one board.

The magnet board, made digital. Drag people onto missions, leave or standby, and everyone sees the same plan, live.

Drag and drop planning
One board per site, one card per person

Where to go, what to wear
The host name opens Google Maps; PPE is on the card

Lock a finished plan
Nobody bumps it by accident



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04 · Manpower Board

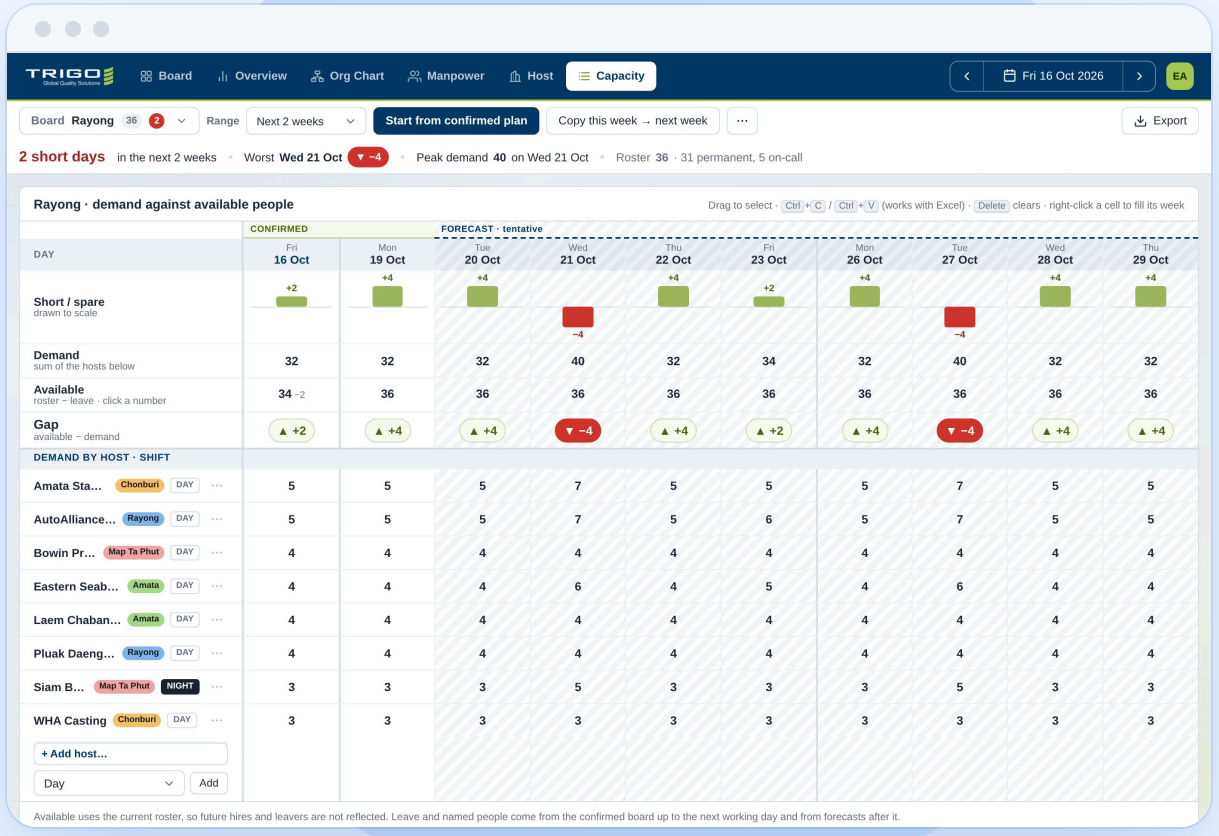
Plan weeks ahead, not just tomorrow.

Carry over the last working day
One click, then change only what moved

Forecast days
Tentative plans kept apart, merged with a side-by-side review

Holds
Take someone another engineer planned and they are told

Export
JPG, PDF and Excel, plus the org chart as PowerPoint



FOR ENGINEERS

Enter the people each host needs per day. The grid shows every short day as you type.

FOR MANAGEMENT

Short days show in red weeks before they happen, with the worst day and peak demand on top.

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04 · Manpower Board

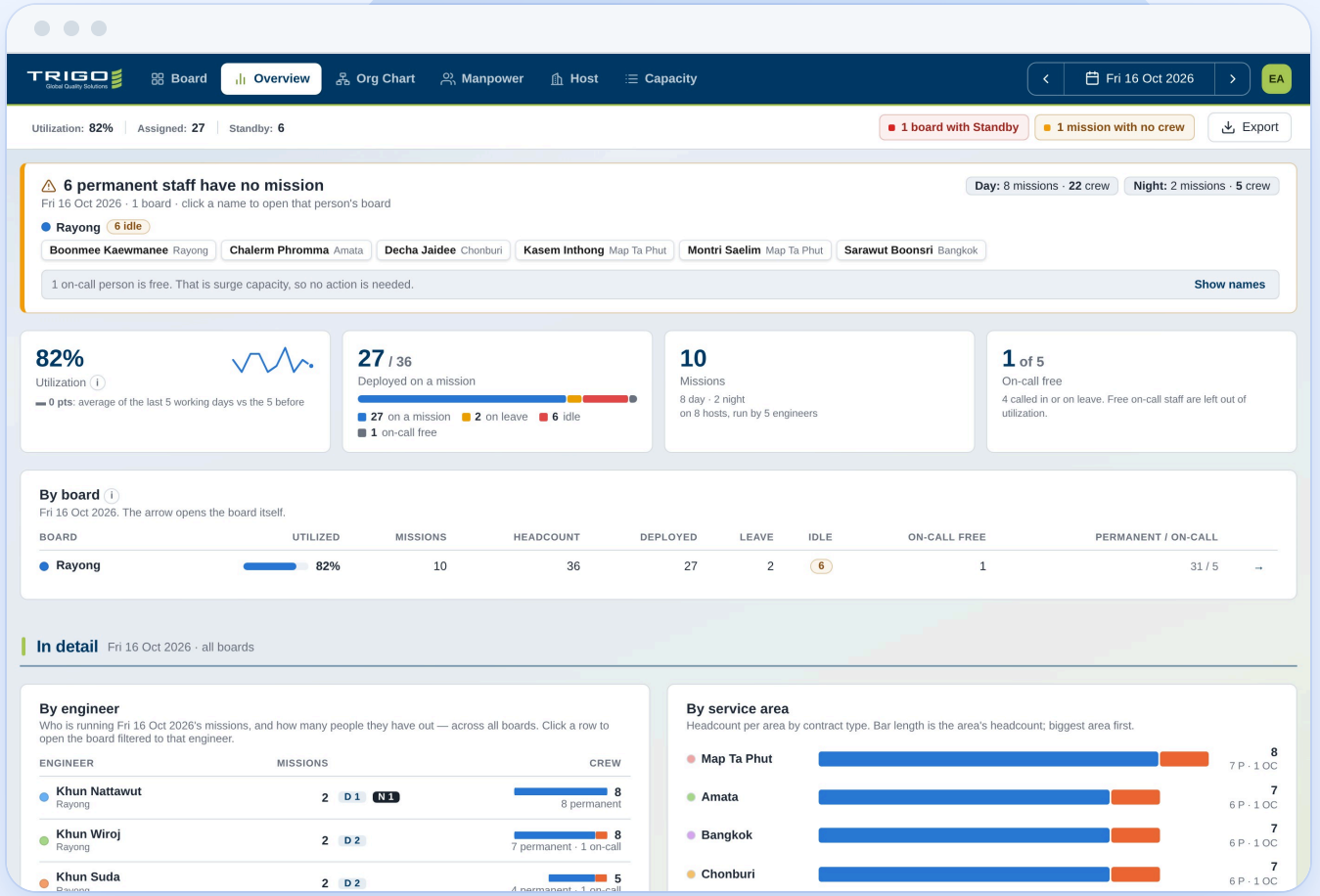
The whole crew in one view.

Utilization you can trust
On-call staff who were not called stay out of the figure

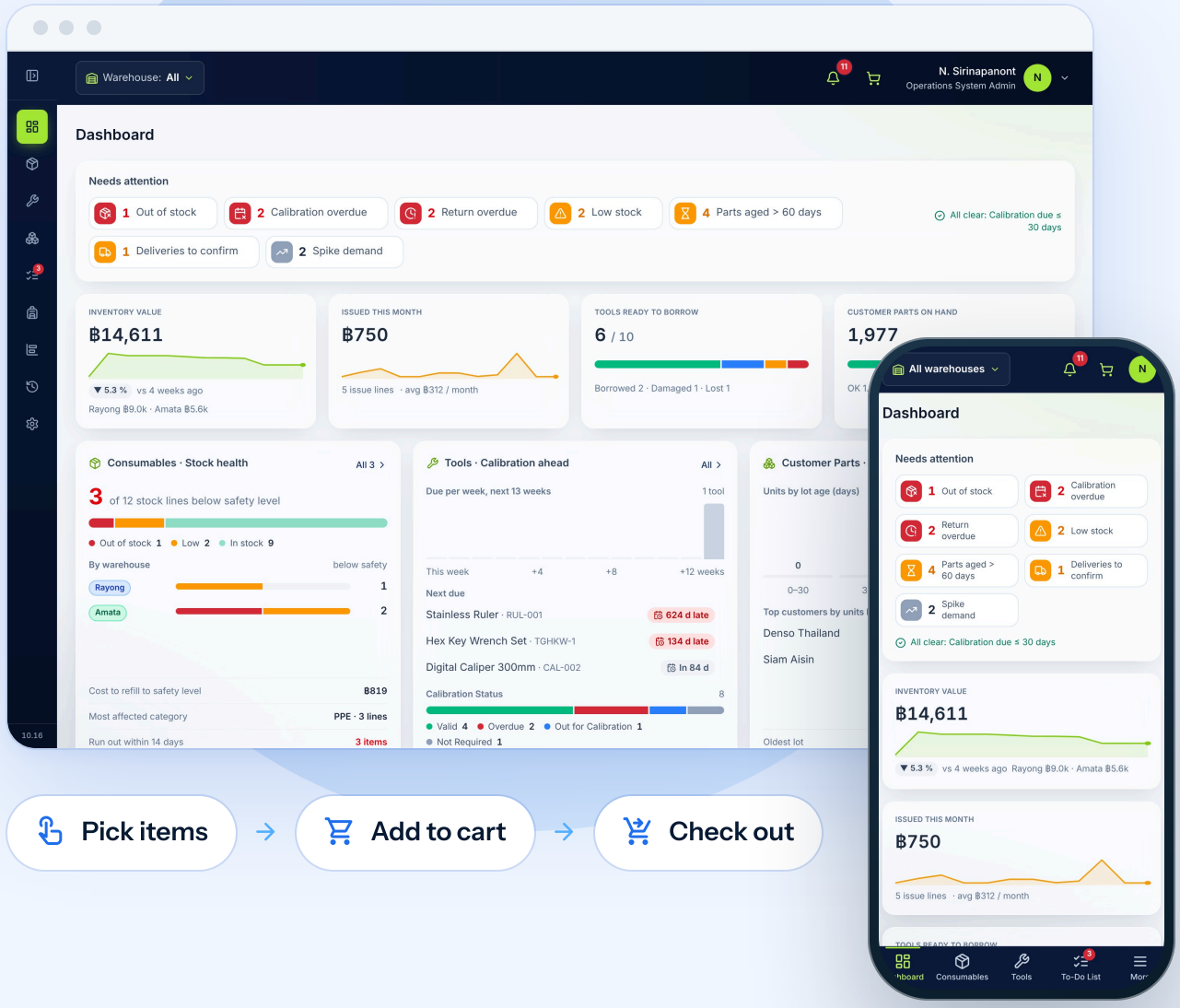
Idle staff named
Permanent staff with no mission, listed by board

By board
Missions, headcount, leave and idle side by side

History and workload
Utilization over time and missions per engineer



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Pick items → Add to cart → Check out

Inventory & Quality System

04 · App 2 of 2

Every tool, part and consumable, accounted for.

Consumables
Stock per warehouse against its safety level

Tools & Calibration
Borrow, return and calibration due dates

Customer Parts
OK, NG, Sorting and Hold per lot

To-Do and Audit Trail
Approvals in one queue, every change on record

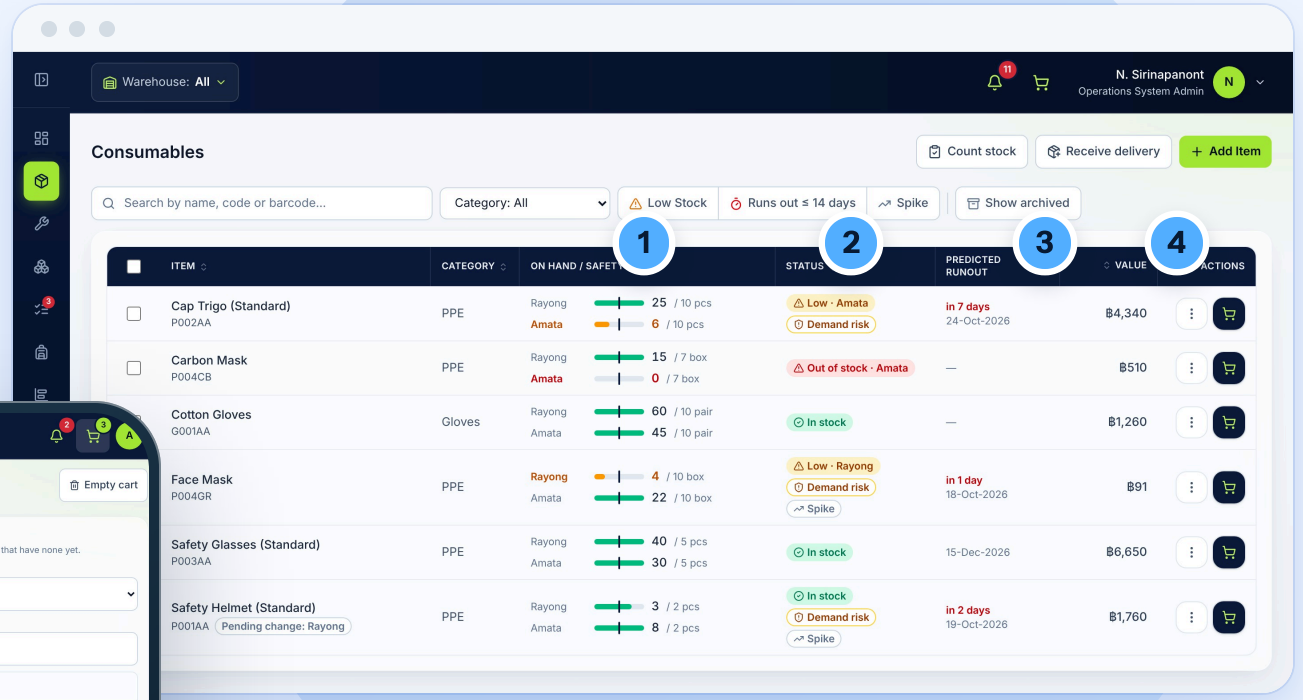
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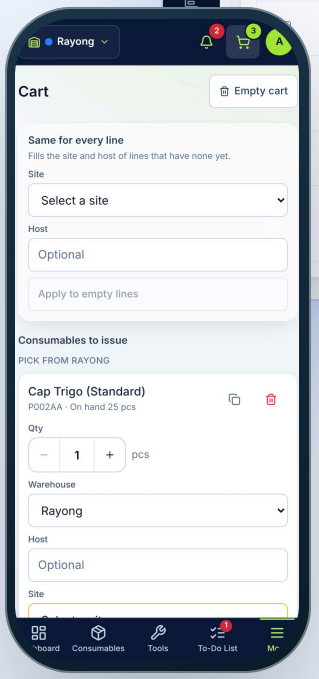
04 · Consumables

Stay above safety stock in every warehouse.

1. **Safety stock per warehouse**
   On hand against the safety level, with a gauge
2. **Clear stock status**
   Out of stock, Low or In stock, warehouse named
3. **Predicted run-out and spikes**
   Days left and demand spikes, before it hurts
4. **Cart for every issue**
   Add items; nothing is sent until you check out

Receive delivery · Count stock · Stock filters





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04 · Inventory & Quality

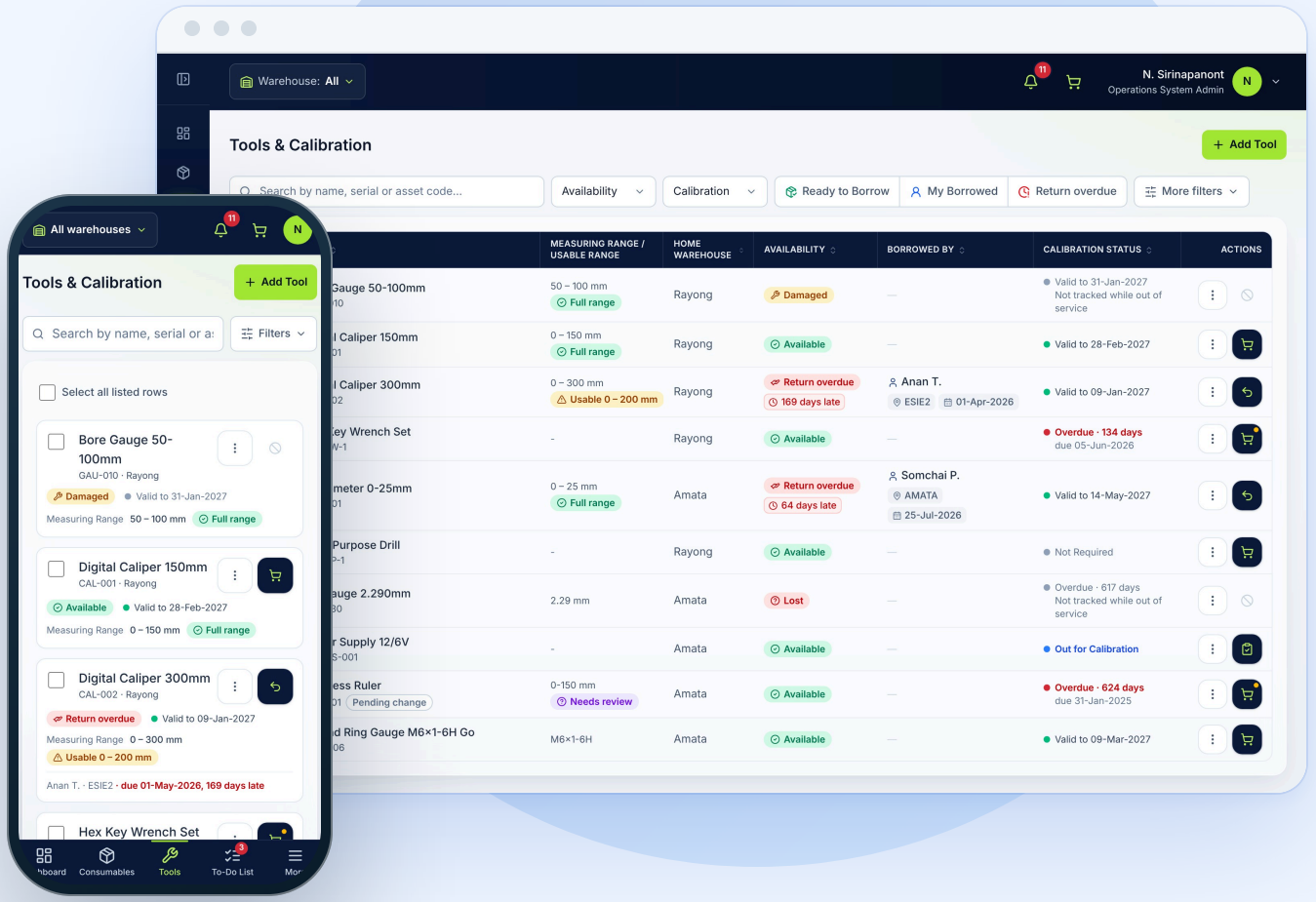
Borrow a tool in a minute.

1 Find it
Search, or scan the barcode with your phone camera

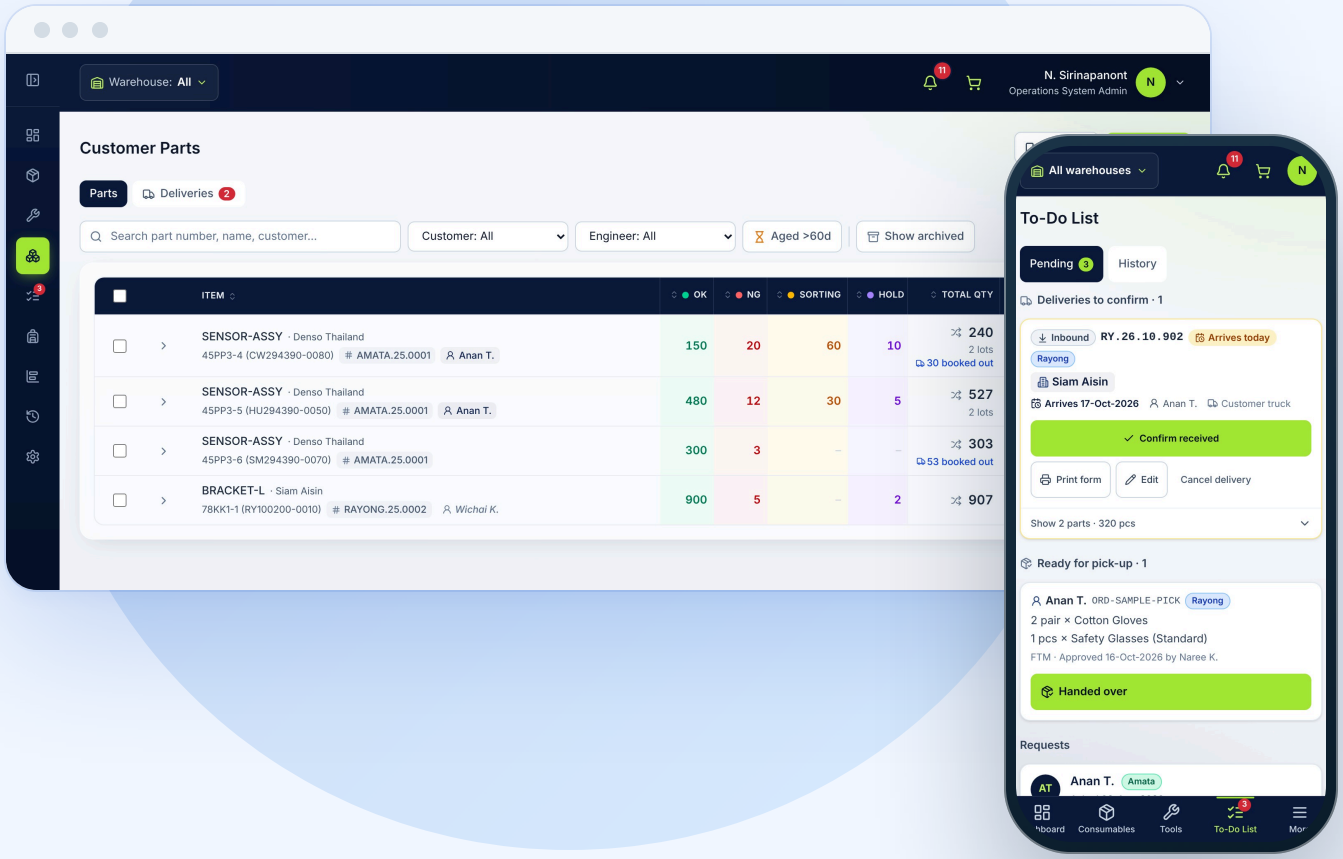
2 Check it
Calibration valid-to date and usable range on every tool

3 Request it
Add to the cart and send one request

4 Return it
Overdue returns are flagged, so nothing goes missing



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04 · Inventory & Quality

Customer parts, lot by lot.

Inspection status per lot
OK, NG, Sorting and Hold, with the total and the lots

Aged parts flagged
Amber past the aged threshold, red past twice it

Deliveries in and out
Several parts on one Asset In-Out Form; booked stock is reserved

Confirm what really arrived
Actual quantities per line; any difference goes in the log

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04 · On your phone

Both apps fit in your pocket.

Each app has its own phone layout and installs from the browser like a native app. No app store needed.

01 **Scan** the code and sign in with your @trigo-group.com email.

02 **iPhone:** in Safari tap Share, then Add to Home Screen.

03 **Android:** in Chrome tap the menu, then Install app.

Manpower Board

trigoth-manpowerboard.netlify.app

Inventory & Quality

trigoth-inventory.pages.dev

Installable app (PWA)

Phone layout